

BUSINESS ANALYTICS CASE STUDY

Bank Marketing: Understanding Term-Deposit Subscription

Using customer and campaign data to identify high-value segments, key subscription drivers, and opportunities for better targeting.

PREPARED BY

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TOOLS

Python · Pandas · Matplotlib · Logistic Regression

BUSINESS PROBLEM

The Business Question

The bank wants to improve the effectiveness of its term-deposit marketing campaigns.

1

Why do some customers subscribe to term deposits while others do not?

2

Which customer groups are more likely to respond positively to marketing campaigns?

BUSINESS OBJECTIVE

Use customer and campaign data to identify high-potential customers and improve campaign targeting.

Understanding the Data

11,162 customers · 17 variables

SUBSCRIBERS

5,289

47.38%

NON-SUBSCRIBERS

5,873

52.62%

VARIABLES ANALYSED

- **Customer Profile**
Age · Job · Education · Marital Status
- **Financial**
Balance · Housing Loan · Personal Loan
- **Campaign**
Contact Method · Call Duration · Campaign Contacts · Previous Outcome

ANALYSIS APPROACH



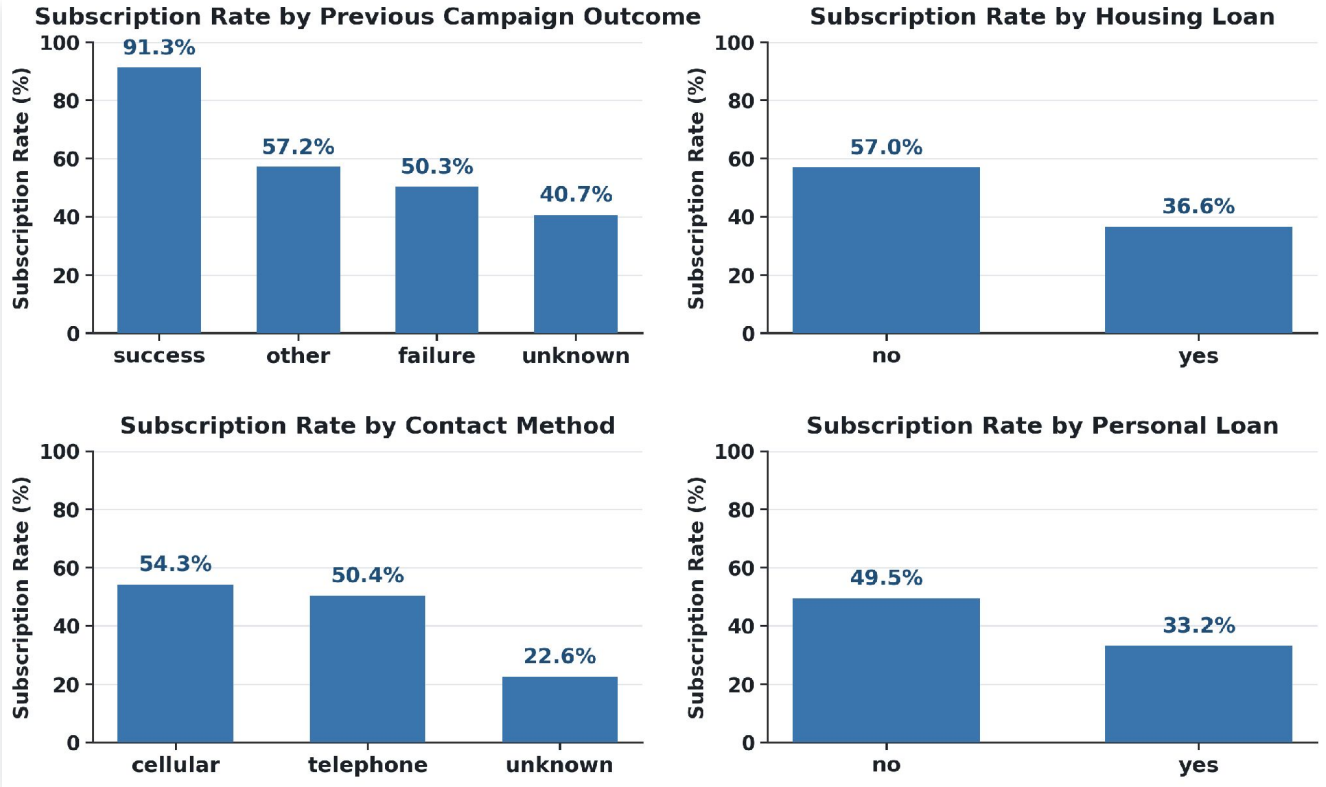
QUESTION 1

Why Do Some Customers Subscribe While Others Do Not?

Some customers are more likely to subscribe because they have positive previous campaign experiences, fewer financial commitments, and reliable contact channels. Customers with weaker or unknown campaign history, existing loans, or unknown contact information are less likely to subscribe.

ADDITIONAL EVIDENCE

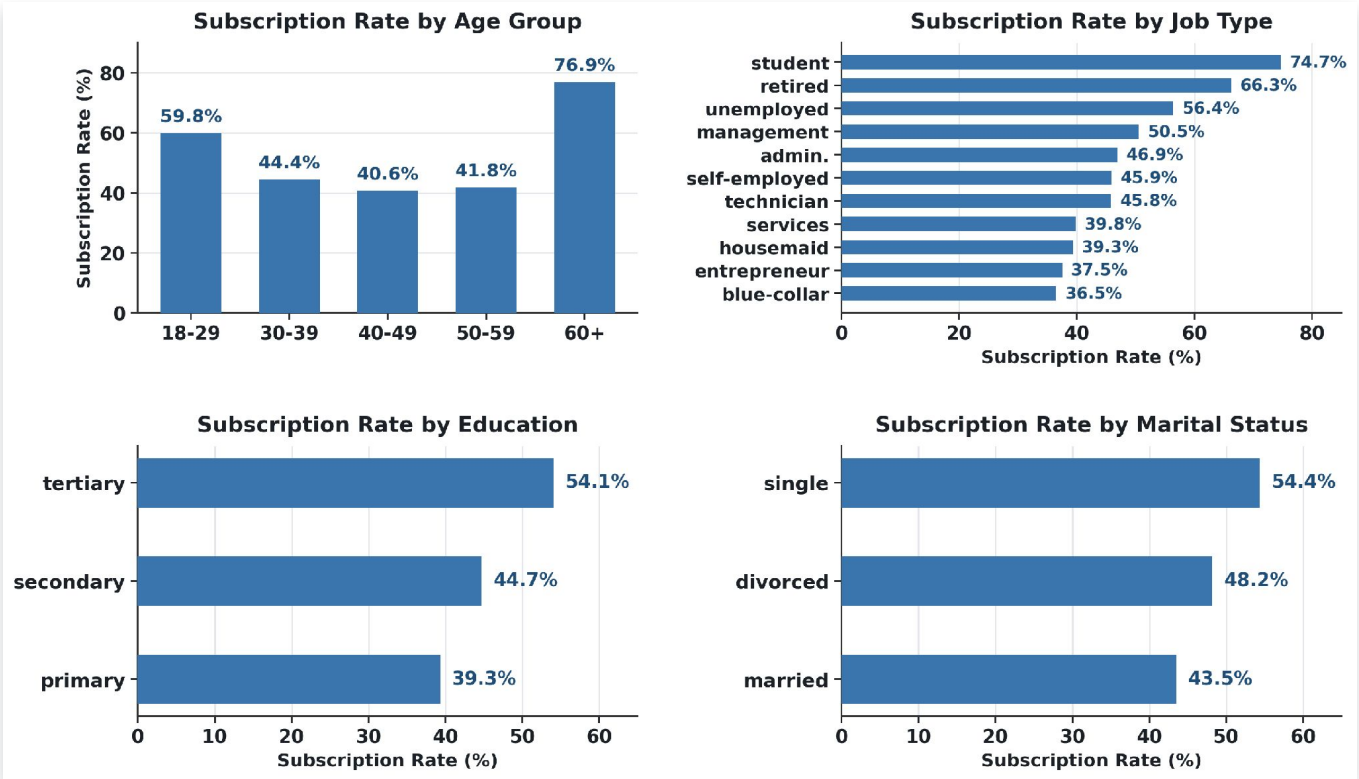
Subscribers also had a higher average balance (1,804 vs 1,280), longer calls (537 vs 223 sec), and fewer campaign contacts (2.14 vs 2.84).



Subscription rate by previous campaign outcome, housing loan, contact method and personal loan

QUESTION 2

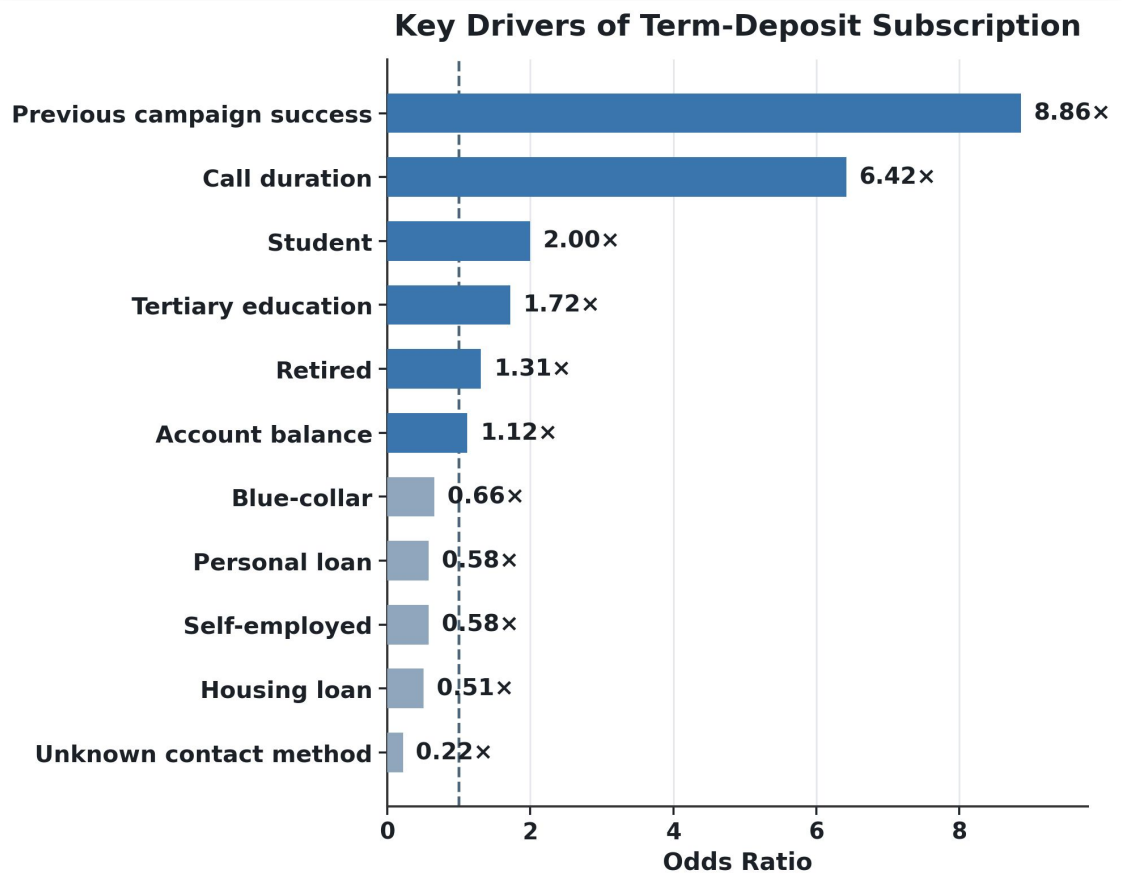
Which Customers Respond More?



BUSINESS TAKEAWAY

The strongest customer segments are older customers, younger customers, students and retirees, plus tertiary-educated (54.1%) and single (54.4%) customers. These groups present the strongest opportunities for targeted campaigns. **The data shows that response rates are highest among customers aged 60+ (76.9%) and 18–29 (59.8%). By occupation, students (74.7%) and retirees (66.3%) are the strongest responding groups. Tertiary-educated customers (54.1%) and single customers (54.4%) also show higher subscription rates within their respective categories.**

What Factors Most Influence Subscription?



Odds ratios from the logistic regression model; the dashed line marks the 1.0x baseline.

KEY TAKEAWAY

Previous campaign success is the strongest predictor of subscription, while customer engagement and selected customer characteristics further help distinguish likely subscribers.

Strongest positive driver

Previous campaign success: 8.86x higher odds of subscribing

What Does the Analysis Tell Us?

01

Previous response is the strongest signal

Customers who previously responded successfully had a 91.3% subscription rate and about 8.9× higher odds of subscribing.

02

Customer engagement matters

Subscribers had an average call duration of 537 seconds, compared with 223 seconds for non-subscribers.

03

Some customer segments respond better

Customers aged 60+ had a 76.9% subscription rate, while students recorded 74.7%.

04

Financial commitments may reduce conversion

Customers without housing loans had a 57.0% subscription rate, compared with 36.6% among customers with housing loans.

05

The model provides strong predictive value

ROC-AUC = 0.907

Strong at distinguishing likely subscribers from non-subscribers

What Should the Bank Take From the Analysis?



Prioritize customers with positive campaign history

Previous success drove a 91.3% subscription rate, versus 40.7% when the previous outcome was unknown. It is the strongest practical signal of a receptive customer.



Focus first on high-response segments

Customers aged 60+ (76.9%), students (74.7%) and retirees (66.3%) should receive greater attention when allocating campaign resources.



Financial commitments may reduce willingness

No housing loan: 57.0% vs 36.6% with one. No personal loan: 49.5% vs 33.2% with one. These customers may need tailored offers.



Reachability and engagement matter

Cellular contact reached 54.3% of customers versus 22.6% for unknown contact methods, with longer average calls (537 vs 223 sec).



Move from broad targeting to prioritization

The model achieved an ROC-AUC of 0.907. Customer scores can rank prospects so campaign effort goes to the most promising contacts first.

OVERALL BUSINESS INSIGHT

The biggest opportunity is not simply contacting more customers, but identifying who is likely to respond and allocating campaign effort accordingly.

Turning Insights Into Action

1	Prioritize customers with previous campaign success	Previous successful responders had a 91.3% subscription rate. Give this group priority in future campaigns.
2	Prioritize high-response customer segments	Allocate more campaign attention to 60+ customers (76.9%), students (74.7%) and retirees (66.3%).
3	Improve customer contact and engagement	Cellular contact reached 54.3% of customers vs 22.6% for unknown methods. Improve contact-data quality and train teams for more meaningful conversations.
4	Develop tailored approaches for customers with financial commitments	Housing-loan customers: 36.6% vs 57.0% without. Personal-loan customers: 33.2% vs 49.5% without. Test offers that better fit these circumstances.
5	Use predictive scoring to prioritize campaign resources	The model achieved an ROC-AUC of 0.907. Use customer scores to rank prospects and contact higher-probability customers first.

CONCLUSION

From Mass Marketing to Smarter Targeting

Term-deposit subscription is shaped by customer history, financial circumstances, campaign engagement and customer characteristics.

Prioritize proven responders

Previous campaign success: 91.3%

Focus on high-response segments

60+: 76.9% · Students: 74.7%

Improve reach and engagement

Cellular: 54.3% vs Unknown: 22.6%

Tailor approaches for lower-response groups

Housing loan: 36.6% · Personal loan: 33.2%

Use data to prioritize prospects

Model ROC-AUC: 0.907

“The opportunity is not to market to everyone equally. It is to identify who is most likely to respond, understand what differentiates them, and use that insight to make campaign resources more effective.”